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## MA 600: Research Methodology

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| Course Code      | MA 600               |
| Course Name      | Research Methodology |
| L-T-P-C          | 1-0-0-1              |
| Prerequisites    | None                 |
| Intended for     | M.Sc/Ph.D.           |
| Mutual Exclusion | RM of other schools  |
| Approval         | 57th BoA (Oct 2024)  |

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### Course Contents

- **Module 1: Introduction to research** (2 Lectures)
  - Defining research: Characteristics and objectives
  - Research and the scientific method
  - Various research methodologies:
    - \* Descriptive vs. Analytical research
    - \* Applied vs. Fundamental research o Quantitative vs. Qualitative research
    - \* Conceptual vs. Empirical research
  - The research process:
    - \* Formulating and defining a research problem
    - \* Developing research questions
    - \* Differentiating between research methods and research methodology
- **Module 2: Literature review and hypothesis development** (1 Lectures)
  - Conducting a literature review: o Reviewing concepts and theories and finding the research gaps.
  - Hypothesis development:
    - \* Identifying sources and characteristics of hypotheses
    - \* Understanding the role of hypotheses in research o Methods for testing hypotheses
- **Module 3: Research communication** (2 Lectures)
  - Writing and structuring research papers
  - Preparing and delivering research presentations
  - Crafting abstracts and summaries
  - Group discussion
  - Fear of rejection
- **Module 4: Research ethics** (2 Lectures)
  - Informed consent and ethical considerations
  - Maintaining confidentiality and privacy
  - Avoiding plagiarism and ensuring proper attribution
  - Ethical data use and management
  - Collaboration and ethical practices in joint research
- **Module 5: LaTeX for academic writing** (2 Lectures)
  - Introduction to LaTeX: Basic commands and structure
  - Writing mathematical equations and symbols

- Organizing documents with sections, tables, and figures
- Creating bibliographies and managing citations
- Customizing document formats for research publications
- **Module 6: Technical writing and research documentation** (3 Lectures)
  - Writing research articles
  - Structuring and writing research projects
  - Writing theses and dissertations
  - Authoring books and writing reviews (e.g., book reviews, case reviews)
  - Understanding the criteria for high quality research
- **Module 7: Citation methods and research integrity** (2 Lectures)
  - Citation Techniques:
    - \* Footnotes, endnotes, and in-text citations
    - \* Compiling and formatting bibliographies
  - Ethical considerations in research and citation
  - Proper application of citation rules and guideline

## Laboratory Modules

1. NA

## Text Books

1. Wayne Goddard, Stuart Melville, **Research Methodology: An Introduction**, Juta and Co. Ltd., 2007.
2. Stefan Kottwitz, **LaTeX Beginner's Guide**, Second Edition, Packt Publishing Ltd., 2011.

## References

1. R. Barker Bausell, **Advanced Research Methodology**, Scarecrow Press, 2013.
2. Bill Taylor, Gautam Sinha, Taposh Ghoshal, **Research Methodology: A Guide for Researchers in Management and Social Sciences**, Prentice-Hall of India Private Limited, 2006.
3. Stefan Kottwitz, **LaTeX Graphics with TikZ: A practitioner's guide to drawing 2D and 3D images, diagrams, charts, and plots**, Packt Publishing Ltd. 2023.